

Chemical Safety and HAZMAT Handling Procedures

Reference: 29 CFR 1910.1200 - OSHA HazCom Standard

Hazard Communication (HazCom)

OSHA's Hazard Communication Standard (HCS) requires chemical manufacturers to evaluate hazards and communicate them via Safety Data Sheets (SDS) and labels. The GHS-aligned SDS has 16 sections providing comprehensive hazard information.

Safety Data Sheets (SDS)

SDS sections: 1-Identification, 2-Hazard Identification, 3-Composition, 4-First Aid, 5-Firefighting, 6-Accidental Release, 7-Handling/Storage, 8-Exposure Controls/PPE, 9-Physical/Chemical Properties, 10-Stability/Reactivity, 11-Toxicology, 12-Ecological, 13-Disposal, 14-Transport, 15-Regulatory, 16-Other Information.

Chemical Storage Requirements

Flammable liquids must be stored in approved flammable storage cabinets away from ignition sources. Incompatible chemicals must be segregated: acids from bases, oxidizers from flammables, and toxics from food/beverages. All containers must be labeled with chemical name, hazard warnings, and date opened.

Chemical Spill Response

Minor spills (under 1 gallon): Don appropriate PPE, contain spill with absorbent materials, clean from outside in, dispose as hazardous waste. Major spills: evacuate area, call emergency response, do not enter without SCBA. Report all spills to supervisor immediately. Spill kit contents: absorbent pads, neutralizing agents, goggles, gloves, waste bags.

Emergency Response Procedures

Chemical emergency contacts: Emergency: 911, CHEMTREC: 1-800-424-9300, Poison Control: 1-800-222-1222. For chemical exposure: remove contaminated clothing, flush with water 15-20 minutes, seek medical attention. Do not induce vomiting for ingested chemicals unless directed by Poison Control.

HAZMAT Classification

DOT hazmat classes: Class 1-Explosives, Class 2-Gases, Class 3-Flammable Liquids, Class 4-Flammable Solids, Class 5-Oxidizers, Class 6-Toxic/Infectious, Class 7-Radioactive, Class 8-Corrosives, Class 9-Miscellaneous. Proper shipping name, UN number, and placards required for transport.

